Deep Reinforcement Learning for Optimal Portfolio Allocation: A Comparative Study with Mean-Variance Optimization

Srijan Sood, [1] **Kassiani Papasotiriou,** [1] **Marius Vaiciulis,**[2] **Tucker Balch** [1]**.**

[1] J.P. Morgan AI Research

[2] J.P. Morgan Global Equities; Oxford-Man Institute of Quantitative Finance

{srijan.sood, kassiani.papasotiriou, marius.vaiciulis, tucker.balch}@jpmorgan.com

Abstract

Portfolio Management is the process of overseeing a group of investments, referred to as a portfolio, with the objective of achieving predetermined investment goals and objectives. Portfolio Optimization is a key component that involves allocating the portfolio assets so as to maximize returns while minimizing risk taken. Portfolio optimization is typically carried out by financial professionals who use a combination of quantitative techniques and investment expertise to make decisions about the portfolio allocation.

Recent applications of Deep Reinforcement Learning (DRL) have shown promising results when used to optimize portfolio allocation by training model-free agents on historical market data. Many of these methods compare their results against basic benchmarks or other state-of-the-art DRL agents but often fail to compare their performance against traditional methods used by financial professionals in practical settings. One of the most commonly used methods for this task is Mean-Variance Portfolio Optimization (MVO), which uses historical timeseries information to estimate expected asset returns and covariances, which are then used to optimize for an investment objective.

Our work is a thorough comparison between model-free DRL and MVO for optimal portfolio allocation. We detail the specifics of how to make DRL for portfolio optimization work in practice, also noting the comparable tweaks needed to get MVO operational. Backtest results display strong improved performance of the DRL agent in terms of many performance metrics, including Sharpe ratio, maximum drawdowns, and absolute returns.

Introduction

Portfolio management is a key issue in the financial services domain. It constitutes allocating funds across a diverse variety of assets, typically to generate uncorrelated returns while minimizing risk and operational costs. Portfolios can constitute holdings across asset classes (cash, bonds, equities, etc.), or can also be optimized within a specific asset class (e.g., picking the appropriate composition of stocks for an equity portfolio). Investors may choose to optimize for various performance criteria, often centered around maximizing portfolio returns relative to risk taken. Since the advent of Modern Portfolio Theory (Markowitz 1952), a lot of progress has been made in both theoretical and applied aspects of portfolio optimization. These range from improvements in the optimization process, to the framing of additional constraints that might be desirable to rational investors (Cornuejols and Tütüncü 2006; Li and Hoi 2014; Kalayci et al. 2017; Ghahtarani, Saif, and Ghasemi 2022). Recently, the community has tapped the many advancements in Machine Learning (ML) to aid with feature selection, forecasting and estimation of asset means and covariances, as well as using gradient based methods for optimization.

Concurrently, the past decade has witnessed the success of Reinforcement Learning (RL) in the fields of gaming, robotics, natural language processing etc. (Silver et al. 2017; Nguyen and La 2019; Su et al. 2016). The sequential decision making nature of Deep RL, along with its success in applied settings, has captured the attention of the finance research community. In particular, some of the most popular areas of focus of the application of DRL in finance have been on automated stock trading (Yang et al. 2020; Théate and Ernst 2021; Zhang, Zohren, and Roberts 2020; Wu et al. 2020), risk management through deep hedging (Buehler et al. 2019; Du et al. 2020; Cao et al. 2021; Benhamou et al. 2020b) and portfolio optimization. In the upcoming section, we'll examine the landscape of DRL in portfolio optimization and trading problems. While these approaches exhibit improved performance than previous studies, they do have some shortcomings. For instance, some generate discrete asset trading signals which limit their use in broader portfolio management. Additionally, majority of these approaches compare results against ML or buy-and-hold baselines, and don't consider classical portfolio optimization techniques, such as Mean-Variance Optimization.

In our work, we aim to compare a simple and robust DRL framework, that was designed around risk-adjusted returns, with one of the traditional finance methods for portfolio optimization, MVO. We train policy gradient based agents on a multi asset trading environment that simulates the US Equities market (using market data replay), and create observation states derived from the observed asset prices. The agents optimize for risk-adjusted returns, not dissimilar to the traditional MVO methods. We compare the performance of the DRL strategy against MVO through a series of systematic backtests, and observe improved performance along many performance metrics, including risk adjusted returns, max

Copyright © 2023, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.